

# Zhuoyan Xu

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## EDUCATION

### University of Wisconsin-Madison

Ph.D. in Statistics (Co-advised by: [Yingyu Liang](#), [Yin Li](#), [Yiqiao Zhong](#))

M.S. in Computer Science

Madison, WI

Sep. 2020 - present

Sep. 2021 - May 2023

### Wuhan University

B.S in Statistics

China

Aug. 2015 – Jun. 2019

## RESEARCH INTERESTS

My research interest mainly focus on Foundation Models (including large vision, language and multimodal models). I am interested in investigating and **analyzing behavior of foundation models**, aiming to **enhance their adaptation** to downstream tasks with improved accuracy and greater efficiency.

## RECENT PUBLICATIONS

- [Zhuoyan Xu](#), Haoyang Fang, Boran Han, Bonan Min, Bernie Wang, Shuai Zhang. [TabRAG: Efficient Table Retrieval and Understanding with Large Multimodal Models](#). Work done during internship at AWS.
- Yingyu Liang\*, Heshan Liu\*, Zhenmei Shi\*, Zhao Song\*, [Zhuoyan Xu\\*](#), Junze Yin\* [Conv-Basis: A New Paradigm for Efficient Attention Inference and Gradient Computation in Transformers](#) arXiv 2024. (\* denotes alphabetical order).
- Jiajun Song, [Zhuoyan Xu](#), Yiqiao Zhong. [Out-of-distribution generalization via composition: a lens through induction heads in Transformers](#). arXiv 2024.
- [Zhuoyan Xu](#), Khoi Duc Nguyen, Preeti Mukherjee, Somali Chatterji, Yingyu Liang, Yin Li. [AdaInf: Adaptive Inference for Resource-Constrained Foundation Models](#). ICML 2024 Workshop.
- [Zhuoyan Xu\\*](#), Zhenmei Shi\*, Yingyu Liang. [Do Large Language Models Have Compositional Ability? An Investigation into Limitations and Scalability](#). COLM 2024. (\* denotes equal contribution).
- Zhenmei Shi, Jenny Wei, [Zhuoyan Xu](#), Yingyu Liang. [Why Larger Language Models Do In-context Learning Differently?](#) ICML 2024.
- [Zhuoyan Xu](#), Zhenmei Shi, Jenny Wei, Fangzhou Mu, Yin Li, Yingyu Liang. [Towards Few-Shot Adaptation of Foundation Models via Multitask Finetuning](#). ICLR 2024.

## PROFESSIONAL EXPERIENCE

### Research Scientist Intern

Amazon AWS AI

May 2024 – Aug. 2024

Bellevue, WA

- Retrieval-augmented generation (RAG) for Multimodal Large Language Model
- Propose **TabRAG**, a novel framework that addresses table understanding challenges by directly utilizing table images in both retrieval and generation step
- Experimental validation is conducted using a newly constructed table image dataset from 14 public table understanding dataset, demonstrating the robustness and efficiency of our proposed framework

### Machine Learning Engineer Intern

John Deere

May 2022 – Aug. 2022

Fargo, ND

- Developed a deep learning framework for electrical machine winding temperature prediction
- Built a data processing and feature engineering pipeline for 1 million long sequence multivarious time series data.
- Trained machine learning frameworks for time series forecasting, achieved 50%+ improvement than traditional statistical model
- Compiled a 60-page detailed document for future work and knowledge sharing

### Research Assistant

UW - Madison

Sep. 2021 – present

Madison, WI

- Towards understanding and better adaptation of foundation models.

## ACADEMIC SERVICES

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**Conference Reviewer:** NeurIPS 2024, ICML 2024, ICLR 2025, AISTATS 2025

## PREPRINTS

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- [Zhuoyan Xu](#), Kris Sankaran. [Spatial Transcriptomics Dimensionality Reduction using Wavelet Bases](#). F1000 Research.
- [Zhuoyan Xu](#), Jiaxin Hu, Miaoyan Wang. [Generalized tensor regression with covariates on multiple modes](#). arXiv.

## SELECTED RESEARCH EXPERIENCE

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### Adaptation of Foundation Models with Multitask Learning

Feb. 2022 – Sep. 2023

- Implemented multitask fine-tuning strategy to enhance the performance of foundational models on downstream tasks with scarce labeled data.
- Provide a theoretical framework to substantiate the efficacy of the multitask finetuning methodology.
- Proposed a task selection algorithm based on our theoretical conclusion that effectively identifies related finetuning tasks, thereby boosting the model's effectiveness on specific target tasks.
- Applied the multitask finetuning and task selection algorithm across various experiments in vision, NLP, and multimodal models, resulting in a substantial increase in accuracy.

### Adaptive Inference of Vision Foundation Models

Nov. 2023 – Now

- Investigate the fast adaptation of pre-trained foundation models balancing accuracy and latency.
- Calculated FLOPS during model inference and contribute to GitHub repo [flops-counter.pytorch](#) (Star: 2.6k).
- Train a scheduler that actively deactivates certain components of model during inference, reducing deploy time while keeping accuracy.

### Exploring In-Context Learning with Large Language Models

Jul. 2023 – Now

- Analyzing general in-context learning tasks through the lens of in-context exemplar selection, utilizing a similarity-based approach.
- Conducted comprehensive studies on the impact of model scale on in-context learning, applying it to classification tasks with models ranging from 70 million to 70 billion parameters.
- Examining the compositional capabilities in in-context learning by designing and testing both simple and complex tasks in linguistics to assess LLM performance.
- Exploring parameter efficient finetuning (PEFT) on LLM on arithmetic tasks, focusing on assessing performance on out-of-distribution data.

## TECHNICAL SKILLS

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**Languages:** Python, R, Julia, Java, C, SQL, HTML

**Developer Tools:** Git, Linux, AWS, Azure, Docker, Google Cloud Platform, Slurm,  $\text{\LaTeX}$

## SOFTWARE

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- **Xu, Z.**, Sankaran, K., 2022. R package *waveST*: Spatial Transcriptomics Dimensionality Reduction using Wavelet Bases. Published on [GitHub](#). DOI: [10.5281/zenodo.6823315](#)
- **Xu, Z.**, Hu, J. and Wang, M., 2019. R package *Tensorregress*: Generalized tensor regression with covariates on multiple modes. Published on the [Comprehensive R Archive Network](#)
- **Xu, Z.**, Solís-Lemus, C. 2022. Julia Package *HighDimMixedModels.jl*: Fitting mixed-effects models with high dimensional fixed effect variable.